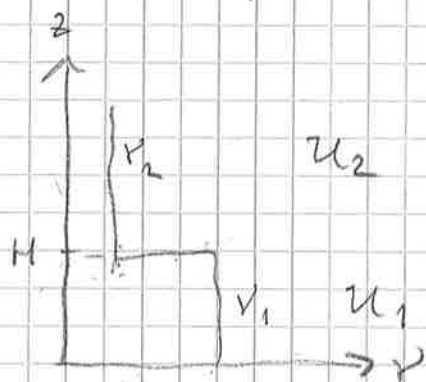


Ekman spiral med två skikt



Ekvation:

$$\frac{d}{dz} \left(\nu \frac{du}{dz} \right) = \epsilon f (u - u_0) \quad (1)$$

$$u = u + i v, \quad u_0 = u_0 \quad (2)$$

Randvillkor:

$$\begin{cases} u_2(0) = 0 \end{cases} \quad (3)$$

$$\begin{cases} u_2 \rightarrow u_0, \quad z \rightarrow \infty \end{cases} \quad (4)$$

Hoppvillkor:

$$\begin{cases} u_1 = u_2 \end{cases} \quad \text{vid } z = H. \quad (5)$$

$$\begin{cases} \nu_1 \frac{du_1}{dz} = \nu_2 \frac{du_2}{dz} \end{cases} \quad (6)$$

Lösningen till (1) för $\nu = \text{const}$ är

$$u = \text{konst} \cdot e^{\pm (1+i)z/h_E} + u_0$$

där

$$h_E = \sqrt{\frac{2\nu}{f}}$$

Lösung:

$$U_1 = A_1 e^{(1+i)z/h_1} + B_1 e^{-(1+i)z/h_1} + u_0 \quad (7)$$

$$U_2 = B_2 e^{-(1+i)z/h_2} + u_0 \quad (8)$$

$\Rightarrow$

$$\frac{\partial U_1}{\partial z} = \frac{1+i}{h_1} \left(A_1 e^{(1+i)z/h_1} - B_1 e^{-(1+i)z/h_1} \right) \quad (9)$$

$$\frac{\partial U_2}{\partial z} = -\frac{1+i}{h_2} B_2 e^{-(1+i)z/h_2} \quad (10)$$

da

$$h_{1,2} = \sqrt{\frac{2\gamma_{1,2}}{\rho}} \quad (12)$$

(3) $\Rightarrow$

$$A_1 + B_1 + u_0 = 0 \quad (13)$$

(5) $\Rightarrow$

$$A_1 E_1 + \frac{B_1}{E_1} = \frac{B_2}{E_2} \quad (14)$$

(6) $\Rightarrow$

$$\frac{\gamma_1}{h_1} (1+i) \left(A_1 E_1 - \frac{B_1}{E_1} \right) = -\frac{\gamma_2}{h_2} (1+i) \frac{B_2}{E_2} \Rightarrow$$

$$A_1 E_1 - \frac{B_1}{E_1} = -\frac{h_1}{h_2} \frac{\gamma_2}{\gamma_1} \frac{B_2}{E_2} = -\frac{h_2}{h_1} \frac{B_2}{E_2} \Rightarrow$$

$$A_1 E_1 - \frac{B_1}{E_1} = -\varepsilon \frac{B_2}{E_2} \quad (15)$$

da

(16)

$$E_1 = e^{(1+\varepsilon)H/h_1} \quad (17)$$

$$E_2 = e^{(1+\varepsilon)H/h_2} \quad (18)$$

$$\xi = \frac{h_2}{h_1} = \sqrt{\frac{\gamma_2}{\gamma_1}} \quad (19)$$

Eliminera B_2/E_2 från (14), (15):

$$A_1 E_1 - \frac{B_1}{E_1} = -\varepsilon \left(A_1 E_1 + \frac{B_1}{E_1} \right) \Rightarrow$$

$$A_1 E_1 (1+\varepsilon) = \frac{B_1}{E_1} (1-\varepsilon) \quad (20)$$

$$(13) \text{ i } (20) \Rightarrow$$

$$-(B_1 + u_0) E_1^2 (1+\varepsilon) = B_1 (1-\varepsilon) \Rightarrow$$

$$B_1 \left[1 - \varepsilon + E_1^2 (1+\varepsilon) \right] = -u_0 E_1^2 (1+\varepsilon) \Rightarrow$$

$$B_1 = -u_0 \frac{E_1^2 (1+\varepsilon)}{1 - \varepsilon + E_1^2 (1+\varepsilon)} \quad (21)$$

$$(21) \text{ och } (22) \Rightarrow$$

$$A_1 = \frac{B_1}{E_1^2} \frac{1-\varepsilon}{1+\varepsilon} = -u_0 \frac{1-\varepsilon}{1 - \varepsilon + E_1^2 (1+\varepsilon)} \quad (22)$$

$$(14) \Rightarrow$$

$$\frac{B_2}{E_2} = -u_0 \frac{E_1 (1-\varepsilon) + E_1 (1+\varepsilon)}{1 - \varepsilon + E_1^2 (1+\varepsilon)} \Rightarrow$$

$$B_2 = -u_0 \frac{2 E_1 E_2}{1 - \varepsilon + E_1^2 (1+\varepsilon)} \quad (23)$$

Kontroller:

1) $n_1 = n_2 \Leftrightarrow E_1 = E_2 \Leftrightarrow \varepsilon = 1$:

$$\begin{cases} A_1 = 0 \\ B_1 = -U_0 \\ B_2 = -U_0 \end{cases} \quad \text{OK}$$

2) $H \rightarrow 0 \Rightarrow E_1 = E_2 = 1 \Rightarrow$

$$\begin{cases} A_1 = -U_0 \frac{1-\varepsilon}{1-\varepsilon+1+\varepsilon} = -\frac{U_0}{2} (1-\varepsilon) \\ B_1 = -U_0 \frac{1+\varepsilon}{1-\varepsilon+1+\varepsilon} = -\frac{U_0}{2} (1+\varepsilon) \\ B_2 = -U_0 \end{cases}$$

$$A_1 + B_1 + U_0 = 0 \quad \text{O.K.}$$

3) $H \rightarrow \infty \Rightarrow E_1 \rightarrow \infty \Rightarrow$

$$\begin{cases} A_1 = 0 \\ B_1 = -U_0 \\ B_2 = 0 \end{cases} \quad \text{OK}$$

Bestäm dU_1/dz vid $z=0$. (9) $\Rightarrow$

$$\left. \frac{\partial U_1}{\partial z} \right|_{z=0} = \frac{1-i}{n_1} (A_1 - B_1) \quad (24)$$

$$(22), (21) \Rightarrow$$

$$A_1 - B_1 = u_0 \frac{E_1^2(1+\varepsilon) - (1-\varepsilon)}{1-\varepsilon + E_1^2(1+\varepsilon)} \quad (25)$$

Om $\varepsilon = 1$ så är $A_1 - B_1$ reellt, och vinkeln vriden 45° . Detta gäller om $H/h_1 \rightarrow \infty \Rightarrow E_1^2 \rightarrow \infty$, och om $H/h_1 \rightarrow 0 \Rightarrow E_1^2 = 1$.

Skriv om (25):

$$A_1 - B_1 = u_0 \frac{E_1^2 - \frac{1-\varepsilon}{1+\varepsilon}}{E_1^2 + \frac{1-\varepsilon}{1+\varepsilon}} = u_0 \frac{E_1^2 - R}{E_1^2 + R} \quad (26)$$

där

$$R = \frac{1-\varepsilon}{1+\varepsilon} \quad (27)$$

Sätt

$$\varphi = \frac{H}{h_1} \quad (28)$$

Antag $\varphi \ll 1, \Rightarrow$

$$E_1 = e^{(1+i)\varphi} \approx 1 + (1+i)\varphi + \frac{1}{2}(1+i)^2\varphi^2 \quad (29)$$

(29) i (26) $\Rightarrow$

$$\begin{aligned} A_1 - B_1 &\approx u_0 \frac{1 + 2(1+i)\varphi - R}{1 + 2(1+i)\varphi + R} = \\ &= u_0 \frac{1-R}{1+R} \frac{1 + 2\frac{1-i}{1-R}\varphi}{1 + 2\frac{1+i}{1+R}\varphi} \approx u_0 \frac{1-R}{1+R} \left[1 + 2(1+i)\varphi \left(\frac{1}{1-R} - \frac{1}{1+R} \right) \right] \end{aligned}$$

Vi har

$$1-R = 1 - \frac{1-\varepsilon}{1+\varepsilon} = \frac{2\varepsilon}{1+\varepsilon}$$

$$1+R = 1 + \frac{1-\varepsilon}{1+\varepsilon} = \frac{2}{1+\varepsilon}$$

$$\frac{1}{1-R} - \frac{1}{1+R} = \frac{1+\varepsilon}{2\varepsilon} - \frac{1+\varepsilon}{2} = \frac{1+\varepsilon}{2} \left(\frac{1}{\varepsilon} - 1 \right) = \frac{1-\varepsilon^2}{2\varepsilon}$$

$$\frac{1-R}{1+R} = \varepsilon$$

$$A_1 - E_1 \approx u_0 \varepsilon \left[1 + 2\varphi(1+i) \frac{1-\varepsilon^2}{2\varepsilon} \right] \quad (30)$$

Antag att $\varepsilon \ll 1$, och $\varphi \ll 1$, men att vi inte känner ε/φ . (27) $\Rightarrow$

$$R \approx 1 - 2\varepsilon \quad (31)$$

(29) och (31) i (26) $\Rightarrow$

$$\begin{aligned} A_1 - E_1 &\approx u_0 \frac{1 + 2(1+i)\varphi - 1 + 2\varepsilon}{1 + 2(1+i)\varphi + 1 - 2\varepsilon} \\ &\approx u_0 [(1+i)\varphi + \varepsilon] \Rightarrow \end{aligned} \quad (32)$$

$$\frac{\partial u_1}{\partial \varepsilon} \bigg|_{\varepsilon=0} = u_0 \frac{1+i}{u_1} [(1+i)\varphi + \varepsilon] \quad (33)$$

vi ser att vinkeln vider mer än 45° .

Bestäm $u(H)$. (7) $\Rightarrow$

$$u(H) = u_1(H) = A_1 E_1 + \frac{B_1}{E_1} + u_0 = [(21), (22)] =$$

$$= -u_0 \frac{E_1(1-\varepsilon) + E_1(1+\varepsilon)}{1-\varepsilon + E_1^2(1+\varepsilon)} + u_0 =$$

$$= u_0 \left(1 - \frac{2E_1}{1-\varepsilon + E_1^2(1+\varepsilon)} \right) =$$

$$= u_0 \frac{1-\varepsilon + E_1^2 + 2E_1^2 - 2E_1}{1-\varepsilon + E_1^2(1+\varepsilon)} \Rightarrow$$

$$U(H) = U_0 \frac{\varepsilon(E_1^2 - 1) + (E_1 - 1)^2}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} =$$

$$= U_0 (E_1 - 1) \frac{\varepsilon(E_1 + 1) + E_1 - 1}{E_1^2 + 1 + \varepsilon(E_1^2 - 1)} \quad (33)$$

För $\varepsilon \ll 1$ och $\varphi \ll 1$ får vi

$$\begin{aligned} U(H) &\approx U_0 (1+i)\varphi \frac{(1+i)\varphi + 2\varepsilon}{2} = \\ &= U_0 (i\varphi^2 + (1+i)\varepsilon\varphi) \end{aligned} \quad (34)$$

Kontrollera att (34) stämmer med en
Taylorutveckling av $U(z)$ utvärderad på
(33). Vi har

$$U(H) \approx \left. \frac{dU}{dz} \right|_{z=0} H + \frac{d^2U}{dz^2} \frac{H^2}{2} + \dots \quad (35)$$

(1) $\Rightarrow$

$$\left. \frac{d^2U}{dz^2} \right|_{z=0} = i \frac{f}{Y_1} (-U_0) = -i U_0 \frac{2}{h_1^2} \quad (36)$$

(33) och (36) i (35) $\Rightarrow$

$$\begin{aligned} U(H) &\approx U_0 (1+i)\varphi [(1+i)\varphi + \varepsilon] - i U_0 \varphi^2 = \\ &= U_0 \{ [(1+i)^2 - i] \varphi^2 + (1+i)\varepsilon\varphi \} = \\ &= U_0 (i\varphi^2 + (1+i)\varepsilon\varphi) = \\ &= U_0 (1+i)\varphi \left[\frac{1}{2}(1+i)\varphi + \varepsilon \right] \end{aligned} \quad (37)$$

OK